

Asymptotics of OEIS A277364

Bell-number growth and a parity-sensitive omitted tail

A mathematical analysis prepared for Vladimir Reshetnikov

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Abstract

The sequence A277364 counts partitions of an n -element set into at most $\lfloor n/2 \rfloor$ nonempty blocks. We prove that this restriction removes an asymptotically negligible proportion of all set partitions, and hence that the sequence has the full Bell-number asymptotic expansion at every algebraic relative-error scale. Writing $r = W_0(n)$, its leading equivalent is

$$a(n) \sim \frac{\exp\{n(r - 1 + r^{-1}) - 1\}}{\sqrt{1 + r}}.$$

We give a self-contained saddle-point proof, the explicit first correction, and an elementary proof that truncation is harmless. We also determine the much smaller difference $B_n - a(n)$: it is asymptotic to the first omitted Stirling number and has a parity-dependent prefactor multiplying $(C\sqrt{n})^n$, where $C = 1.0658863584\dots$. The omitted proportion is $\exp\{-\frac{1}{2}n \log n + n \log \log n + O(n)\}$. An explicit first correction for this tail, a fixed-fraction generalization, and reproducible exact-integer computations complete the analysis.

1 Definition and principal conclusions

The OEIS entry defines A277364 by the finite Stirling sum

$$a(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} S(n, k), \quad S(n, k) = \begin{Bmatrix} n \\ k \end{Bmatrix}. \quad (1.1)$$

The entry was contributed by Vladimir Reshetnikov in 2016. Its displayed terms begin

$$1, 0, 1, 1, 8, 16, 122, 365, 2795, 11051, 86472, 422005, \dots$$

The definition and these initial values are taken from the retrieved OEIS entry [1]; the calculations below independently reproduce all 26 values displayed there. In particular, $a(0) = 1$ and $a(1) = 0$.

Let

$$B_n = \sum_{k=0}^n S(n, k), \quad K_n = \lfloor n/2 \rfloor + 1, \quad D_n = B_n - a(n) = \sum_{k=K_n}^n S(n, k). \quad (1.2)$$

Thus B_n is the n th Bell number. Our convention is that the blocks are nonempty and unordered, and the elements are labeled, as in the standard Bell and Stirling definitions [3, 4].

Remark 1.1 (What is, and is not, being restricted). The bound concerns the *number of blocks*, not their individual sizes. Singleton blocks are allowed. For example, $\{\{1\}, \{2, 3, 4\}\}$ is counted when $n = 4$. The condition says that the average block size is at least two, not that every block has size at least two.

Throughout the article, logarithms are natural unless a base is specified. The notation $f(n) \sim g(n)$ means that $f(n)/g(n) \rightarrow 1$. Let W_0 denote the principal real Lambert function, so that

$$r = r_n = W_0(n) > 0, \quad re^r = n. \quad (1.3)$$

Define

$$F(n) = \frac{\exp\{n(r - 1 + r^{-1}) - 1\}}{\sqrt{1+r}}, \quad (1.4)$$

$$P(r) = \frac{r^2(2r^2 + 7r + 10)}{24(1+r)^3}. \quad (1.5)$$

Theorem 1.2 (Asymptotic growth). *As $n \rightarrow \infty$,*

$$a(n) = F(n) \left(1 - \frac{P(r)}{n} + O\left(\frac{r^2}{n^2}\right) \right). \quad (1.6)$$

In particular,

$$a(n) \sim B_n \sim \frac{1}{\sqrt{1+W_0(n)}} \left(\frac{n}{eW_0(n)} \right)^n \exp\left(\frac{n}{W_0(n)} - 1\right). \quad (1.7)$$

Consequently, on the coarser logarithmic and root scales,

$$\log a(n) = n \log n - n \log \log n - n + o(n), \quad (1.8)$$

$$a(n)^{1/n} \sim \frac{n}{e \log n}. \quad (1.9)$$

The Bell-number factor in this theorem is classical. Moser and Wyman obtained a full saddle-point expansion [6]; NIST records the leading equivalent in terms of the solution of $N \log N = n$ [3]. The task particular to A277364 is to justify discarding the upper half of the Stirling row. We prove this first without using any Bell asymptotic. We subsequently quantify the discarded part itself.

Growth-rate summary. A277364 has Bell-number growth, not the much smaller growth of the boundary term $S(n, \lfloor n/2 \rfloor)$. The number of summands alone is misleading: the substantial mass of the Stirling row lies well to the left of the cutoff. The omitted fraction satisfies

$$\log \frac{B_n - a(n)}{B_n} = -\frac{1}{2}n \log n + n \log \log n + O(n).$$

In particular, it is smaller than e^{-cn} for every fixed $c > 0$, once n is sufficiently large.

2 An elementary proof that the cutoff is negligible

Lemma 2.1 (Two elementary counting bounds). *For $1 \leq k \leq n$,*

$$k^{n-k} \leq S(n, k) \leq \binom{n}{k} k^{n-k}. \quad (2.1)$$

Consequently,

$$D_n \leq 2^n n^{n-K_n} \leq 2^n n^{n/2}. \quad (2.2)$$

Proof. For the lower bound, place elements $1, \dots, k$ in distinct blocks and assign each of the remaining $n - k$ elements to one of those blocks. The resulting partitions are distinct, giving k^{n-k} possibilities.

For the upper bound, encode a partition by its set of k block minima and, for every other element, the minimum of its block. There are at most $\binom{n}{k} k^{n-k}$ such encodings. Some encodings do not correspond to a valid set of minima, which only makes the upper bound more generous. For $k \geq K_n$, we have $k^{n-k} \leq n^{n-K_n}$. Summing over these k and using $\sum_k \binom{n}{k} = 2^n$ gives (2.2). $\square$

Proposition 2.2 (Super-exponentially small omitted proportion). *There is a constant C_0 such that, for all sufficiently large n ,*

$$0 < \frac{D_n}{B_n} \leq \exp\left(-\frac{1}{2}n \log n + n \log \log n + C_0 n\right). \quad (2.3)$$

Hence $a(n)/B_n \rightarrow 1$, and $D_n/B_n = o(n^{-M})$ for every fixed $M > 0$.

Proof. Take $k = \lfloor n/\log n \rfloor$. The lower bound in Lemma 2.1 gives $B_n \geq k^{n-k}$. Since

$$\log k = \log n - \log \log n + O(\log n/n),$$

we have

$$\log B_n \geq (n - k) \log k = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (2.4)$$

Subtract this lower bound from the logarithm of (2.2). The result is (2.3) after increasing C_0 . Its exponent tends to $-\infty$ faster in magnitude than either cn or $M \log n$, for any fixed positive c, M . $\square$

This proof uses no concentration theorem and no asymptotic formula for Stirling numbers. It also explains why every fixed-order relative expansion of B_n in inverse powers of n , with logarithmic factors, transfers unchanged to $a(n)$. The difference is smaller than all such error scales. An asymptotic expansion of $a(n)$ alone therefore conceals the interesting parity effect in D_n .

3 The Bell saddle and its first correction

We now derive the Bell estimate needed for Theorem 1.2. The generating-function identities

$$S(n, k) = \frac{n!}{k!} [z^n] (e^z - 1)^k, \quad \sum_{n \geq 0} B_n \frac{z^n}{n!} = \exp(e^z - 1) \quad (3.1)$$

follow by forming a set of nonempty labeled blocks. They are also recorded in NIST's Bell and Stirling sections [3, 4].

3.1 The saddle-point integral

Cauchy's formula on the circle $z = re^{i\theta}$ gives

$$B_n = \frac{n! \exp(e^r - 1)}{2\pi r^n} \int_{-\pi}^{\pi} \exp H(\theta) d\theta, \quad H(\theta) = e^{re^{i\theta}} - e^r - in\theta. \quad (3.2)$$

The linear term disappears precisely when $re^r = n$. With $D = z d/dz$, the higher angular derivatives are determined by $D^j e^z$. In particular, putting

$$b = n(1 + r), \quad (3.3)$$

$$c = n(1 + 3r + r^2), \quad (3.4)$$

$$d = n(1 + 7r + 6r^2 + r^3), \quad (3.5)$$

we obtain locally

$$H(\theta) = -\frac{b}{2}\theta^2 - \frac{ic}{6}\theta^3 + \frac{d}{24}\theta^4 + \dots. \quad (3.6)$$

The Gaussian width is $b^{-1/2} \asymp (nr)^{-1/2}$.

3.2 Why the rest of the contour is negligible

Here are explicit estimates that justify using the local expansion. The real loss in the exponent is

$$-\Re H(\theta) = e^r - e^{r \cos \theta} \cos(r \sin \theta). \quad (3.7)$$

For $|\theta| \leq 1/r$, elementary sine and cosine inequalities give

$$-\Re H(\theta) \geq c_1 nr \theta^2 \quad (3.8)$$

with an absolute $c_1 > 0$, for all sufficiently large r . Indeed, the term $e^{r \cos \theta}(1 - \cos(r \sin \theta))$ alone has this lower bound. For $1/r \leq |\theta| \leq \pi$, discard the cosine factor to obtain

$$\begin{aligned} -\Re H(\theta) &\geq e^r - e^{r \cos \theta} \\ &\geq e^r (1 - \exp\{-r(1 - \cos(1/r))\}) \geq c_2 n/r^2. \end{aligned} \quad (3.9)$$

Thus no second part of the circle contributes on the Gaussian scale.

For completeness, set $\varepsilon = \sqrt{r/n}$ and $u = \sqrt{b} \theta$. For each fixed $j \geq 3$,

$$\frac{D^j e^r}{b^{j/2}} = O(\varepsilon^{j-2}). \quad (3.10)$$

This follows from $D^j e^r = e^r T_j(r)$, where T_j is a degree- j polynomial. On $|u| \leq \varepsilon^{-1/12}$, Taylor expansion of the exponential, with the Gaussian factor kept outside, has an integrable polynomial remainder of order $O(\varepsilon^4)$. The terms of orders ε and ε^3 are odd and imaginary; they integrate to zero over this symmetric interval. Estimate (3.8) handles the remaining interval up to $1/r$, with loss at least a positive multiple of $\varepsilon^{-1/6}$, and (3.9) handles the outer arc. Both errors are smaller than any fixed power of ε . These observations justify the error term in the next calculation, not merely its formal coefficients.

3.3 Gaussian contractions

The first nonzero correction comes from the quartic term and the square of the cubic term. Since the second, fourth, and sixth centered Gaussian moments with unit variance are 1, 3, 15, respectively,

$$B_n = \frac{n! \exp(n/r - 1)}{r^n \sqrt{2\pi n(1+r)}} \left(1 + \frac{d}{8b^2} - \frac{5c^2}{24b^3} + O\left(\frac{r^2}{n^2}\right) \right). \quad (3.11)$$

Substitution of (3.3)–(3.5) simplifies the correction to

$$\frac{d}{8b^2} - \frac{5c^2}{24b^3} = -\frac{2r^4 + 9r^3 + 16r^2 + 6r + 2}{24n(1+r)^3}. \quad (3.12)$$

Finally,

$$n! = \sqrt{2\pi n}(n/e)^n \left(1 + \frac{1}{12n} + O(n^{-2})\right).$$

The $1/(12n)$ term must be included: it changes the correction from (3.12) to $-P(r)/n$. Using $n/r = e^r$ gives

$$B_n = F(n) \left(1 - \frac{P(r)}{n} + O(r^2/n^2)\right). \quad (3.13)$$

Proposition 2.2 now proves (1.6).

Remark 3.1 (Two normalizations). Formula (3.11) leaves $n!$ exact, whereas (3.13) expands it. Their leading terms are equivalent, but their first correction coefficients differ by $1/12$. Mixing the prefactor from one normalization with the correction from the other produces an incorrect first-order formula.

4 Logarithmic growth, ratios, and interpretation

4.1 A logarithmic expansion

Taking logarithms in (1.6) yields the comparatively precise form

$$\log a(n) = n(r - 1 + r^{-1}) - 1 - \frac{1}{2} \log(1 + r) - \frac{P(r)}{n} + O(r^2/n^2). \quad (4.1)$$

Let $L = \log n$ and $\ell = \log \log n$. The expansion obtained by solving $r + \log r = L$ is

$$r = L - \ell + \frac{\ell}{L} + \frac{\ell(\ell - 2)}{2L^2} + \frac{\ell(2\ell^2 - 9\ell + 6)}{6L^3} + O\left(\frac{\ell^4}{L^4}\right). \quad (4.2)$$

This is the positive-real specialization of the standard Lambert expansion [5]. It can also be verified directly by substitution; the derivative of $r + \log r$ is bounded away from zero for large r , so its residual controls the error in r . Combining (4.1) and (4.2) gives

$$\frac{\log a(n)}{n} = L - \ell - 1 + \frac{\ell + 1}{L} + \frac{\ell^2}{2L^2} + \frac{\ell^2(2\ell - 3)}{6L^3} + O\left(\frac{\ell^4}{L^4}\right). \quad (4.3)$$

This proves (1.8) and (1.9).

Remark 4.1 (A distinction essential for asymptotic equivalence). It is correct to write

$$a(n) = \left(\frac{n}{e \log n}\right)^n \exp(o(n)).$$

It is *not* correct to replace this by $a(n) \sim (n/(e \log n))^n$. In fact,

$$\log \frac{a(n)}{(n/(e \log n))^n} \sim \frac{n(\log \log n + 1)}{\log n} \rightarrow +\infty.$$

A small error in $\log a(n)/n$ may be a large multiplicative error in $a(n)$. The Lambert- W expression retains the terms needed for a true relative equivalent. Moser and Wyman explicitly discussed this issue when expanding their saddle parameter [6, §3].

4.2 Consecutive terms and the typical block scale

The leading formula also gives

$$\frac{a(n+1)}{a(n)} \sim \frac{n}{W_0(n)}. \quad (4.4)$$

For example, extend F to positive real arguments. Differentiating $re^r = n$ gives $r' = r/(n(1+r))$, and

$$\frac{d}{dn} \log F(n) = r - \frac{r}{2n(1+r)^2}.$$

Integrating over $[n, n+1]$ and using (1.6) proves (4.4). The same conclusion holds for B_{n+1}/B_n .

If N_n is the number of blocks in a uniformly random partition of an n -element set, the Stirling recurrence gives the exact identity

$$\mathbb{E}N_n = \frac{B_{n+1}}{B_n} - 1 \sim \frac{n}{W_0(n)}. \quad (4.5)$$

Indeed, summing $S(n+1, k) = kS(n, k) + S(n, k-1)$ over k proves the first identity. Markov's inequality consequently gives the weaker, but illuminating, estimate

$$\mathbb{P}(N_n > n/2) \leq \frac{2\mathbb{E}N_n}{n} \sim \frac{2}{W_0(n)} \rightarrow 0.$$

The cutoff $n/2$ is far above the mean scale $n/\log n$. This explains the leading answer, although Markov's bound does not measure the true tail. In particular, a central normal approximation near the mean should not be extrapolated to this far-out cutoff to obtain a sharp estimate.

The growth is faster than every fixed exponential, but much smaller than $n!$: from (4.3) and Stirling's formula,

$$\left(\frac{a(n)}{n!}\right)^{1/n} \sim \frac{1}{\log n}.$$

Thus the ordinary generating series has radius zero, while the exponential generating series of $a(n)$ is entire. These are direct root-test consequences, not additional enumeration hypotheses.

5 A sharp description of the omitted tail

The exact first omitted index is important. Put

$$\delta_n = K_n - \frac{n}{2} = \begin{cases} 1, & n \text{ even,} \\ \frac{1}{2}, & n \text{ odd.} \end{cases} \quad (5.1)$$

Let τ be the unique positive solution of

$$\frac{\tau}{1 - e^{-\tau}} = 2. \quad (5.2)$$

The left side increases strictly from 1 to infinity: its derivative has numerator $1 - (1 + \tau)e^{-\tau} > 0$. This proves existence and uniqueness. Equivalently,

$$\tau = 2 + W_0(-2e^{-2}) = 1.5936242600400400923 \dots \quad (5.3)$$

The other real Lambert branch gives $\tau = 0$, which is not a solution of (5.2); the apparent zero only solves the equation after multiplication by $1 - e^{-\tau}$.

Define the constants

$$\begin{aligned} g &= e^\tau - 1 = 3.9215536345675050925 \dots, \\ v &= 2(\tau - 1) = 1.1872485200800801846 \dots, \\ \rho &= 2g = 7.8431072691350101849 \dots, \\ C &= \frac{\sqrt{2g/e}}{\tau} = 1.0658863584129592646 \dots \end{aligned} \quad (5.4)$$

Here v will be the variance of a tilted block size.

Theorem 5.1 (The first omitted block count controls the tail). *As $n \rightarrow \infty$, along either parity,*

$$D_n = S(n, K_n) \left(1 + \frac{\rho}{n} + O(n^{-2}) \right), \quad (5.5)$$

and

$$D_n = \frac{n!}{K_n!} \frac{g^{K_n}}{\tau^n \sqrt{2\pi K_n v}} (1 + O(n^{-1})). \quad (5.6)$$

In explicit growth form, with $\delta = \delta_n$,

$$D_n = T(n) (1 + O(n^{-1})), \quad T(n) = \frac{\rho^\delta}{\sqrt{\pi(\tau - 1)}} n^{-\delta-1/2} (C\sqrt{n})^n. \quad (5.7)$$

In particular, the two subsequences have the distinct equivalents

$$D_n \sim \begin{cases} 5.7432457235111393944 \dots n^{-3/2} (C\sqrt{n})^n, & n \text{ even}, \\ 2.0507528343072577922 \dots n^{-1} (C\sqrt{n})^n, & n \text{ odd}. \end{cases} \quad (5.8)$$

The tail is absolutely enormous, even though its proportion of B_n is extremely small. The parity dependence affects a power of n , not the leading exponential factor. Replacing K_n by $n/2$ before evaluating the factorials would erase precisely this distinction.

6 Proof by tilted block sizes and a local limit estimate

6.1 An exact probabilistic coefficient identity

For a fixed $t > 0$, let X have the positive-integer distribution

$$\mathbb{P}(X = j) = \frac{t^j}{j!(e^t - 1)}, \quad j = 1, 2, \dots \quad (6.1)$$

This is a Poisson distribution conditioned to be positive. If $X_1, \dots, X_k$ are independent copies and $Z_k = X_1 + \dots + X_k$, then

$$S(n, k) = \frac{n!}{k!} \frac{(e^t - 1)^k}{t^n} \mathbb{P}(Z_k = n). \quad (6.2)$$

To see this, expand the probability generating function $((e^{tz} - 1)/(e^t - 1))^k$ and compare with (3.1). The identity is exact for every $t > 0$; the tilt is selected only to make the probability easy to estimate.

The mean and variance in (6.1) are

$$\mu(t) = \frac{t}{1 - e^{-t}}, \quad v(t) = t\mu'(t) = \mu(t)(1 + t - \mu(t)). \quad (6.3)$$

At $t = \tau$, the mean is 2 and the variance is $v = 2(\tau - 1)$. For $k = K_n$, the desired value n differs from the mean $2K_n$ by $-2\delta_n$, which is either -2 or -1 . Thus we only need a local estimate a bounded distance from the mean of this *tilted* distribution. This is different from using a normal approximation for the number of blocks in an unrestricted random partition.

6.2 A local lemma with its first correction

Lemma 6.1 (Local probability and a uniform maximum bound). *Fix $t > 0$ in (6.1). Let $\mu, v > 0$, κ_3 , and κ_4 be its mean, variance, third cumulant, and fourth cumulant. For bounded real d such that $k\mu + d$ is an integer,*

$$\mathbb{P}(Z_k = k\mu + d) = \frac{1}{\sqrt{2\pi kv}} \left(1 + \frac{E(d)}{k} + O(k^{-2}) \right), \quad (6.4)$$

uniformly over those d , where

$$E(d) = \frac{\kappa_4}{8v^2} - \frac{5\kappa_3^2}{24v^3} - \frac{\kappa_3 d}{2v^2} - \frac{d^2}{2v}. \quad (6.5)$$

Moreover, for some $M_t < \infty$,

$$\sup_{m \in \mathbb{Z}} \mathbb{P}(Z_k = m) \leq M_t k^{-1/2}. \quad (6.6)$$

Proof. The characteristic function is

$$\phi(\theta) = \frac{e^{te^{i\theta}} - 1}{e^t - 1}.$$

Since every positive integer has positive probability, the lattice span is one, and $|\phi(\theta)| < 1$ for $0 < |\theta| \leq \pi$. On each closed interval separated from zero it is bounded by some constant strictly below one. Near zero,

$$\log \phi(\theta) - i\mu\theta = -v\theta^2/2 - i\kappa_3\theta^3/6 + \kappa_4\theta^4/24 + O(\theta^5). \quad (6.7)$$

In particular $|\phi(\theta)| \leq \exp(-c\theta^2)$ near zero. Fourier inversion and these two bounds immediately give (6.6).

For (6.4), use Fourier inversion at $k\mu + d$ and set $\theta = u/\sqrt{k}$. Apart from the factor $\exp(-vu^2/2)$, the terms through order $1/k$ are

$$1 - \frac{i}{\sqrt{k}} \left(du + \frac{\kappa_3 u^3}{6} \right) + \frac{1}{k} \left\{ \frac{\kappa_4 u^4}{24} - \frac{1}{2} \left(du + \frac{\kappa_3 u^3}{6} \right)^2 \right\}.$$

Odd terms integrate to zero. The even Gaussian moments are $1/v$, $3/v^2$, and $15/v^3$, which yield exactly (6.5). For the stated remainder, expand through order $k^{-3/2}$ on, for example, $|u| \leq k^{1/24}$. All terms of order $k^{-3/2}$ are again odd. Analyticity of the cumulant generating function near zero bounds the remaining terms by k^{-2} times an integrable Gaussian-polynomial majorant. Gaussian decay controls the rest of the small arc, and the strict bound $|\phi| < 1$ controls the remote arc. The estimates are uniform when d ranges over a fixed bounded interval. $\square$

6.3 Summing the tail: a uniform justification

It is not enough merely to observe that two neighboring Stirling numbers have a small ratio: the entire rest of the sum must be controlled. Use (6.2) at $t = \tau$ and abbreviate $K = K_n$. For every $j \geq 0$,

$$\frac{S(n, K+j)}{S(n, K)} = g^j \frac{K!}{(K+j)!} \frac{\mathbb{P}(Z_{K+j} = n)}{\mathbb{P}(Z_K = n)}. \quad (6.8)$$

Terms with $K+j > n$ vanish. Lemma 6.1 gives $\mathbb{P}(Z_K = n) \geq c/\sqrt{K}$ for sufficiently large n , and its uniform maximum bound gives, with a constant independent of j ,

$$\frac{S(n, K+j)}{S(n, K)} \leq M \sqrt{\frac{K}{K+j}} g^j \frac{K!}{(K+j)!} \leq M \left(\frac{g}{K+1} \right)^j. \quad (6.9)$$

Therefore the combined contribution of all $j \geq 2$ is $O(n^{-2})$. For $j = 1$, both relevant offsets from the mean are bounded, so (6.4) sharpens (6.8) to

$$\begin{aligned} \frac{S(n, K+1)}{S(n, K)} &= \frac{g}{K+1} \sqrt{\frac{K}{K+1}} (1 + O(K^{-1})) \\ &= \frac{2g}{n} + O(n^{-2}). \end{aligned} \quad (6.10)$$

Together these estimates prove (5.5).

The leading term of (6.4), used for $k = K$, gives (5.6). Stirling's formula then gives

$$\begin{aligned} \frac{n!}{K!} \frac{g^K \tau^{-n}}{\sqrt{2\pi K v}} &\sim \frac{2}{\sqrt{2\pi v n}} \left(\frac{2g}{n} \right)^\delta \left(\frac{\sqrt{2g/e}}{\tau} \sqrt{n} \right)^n \\ &= \frac{\rho^\delta}{\sqrt{\pi(\tau-1)}} n^{-\delta-1/2} (C\sqrt{n})^n. \end{aligned}$$

The relative error in this calculation is $O(n^{-1})$ for bounded δ . This completes the proof of Theorem 5.1.

7 An explicit first correction for the tail

The same calculation supplies a useful refinement of (5.7). For the tilt at τ , the cumulants needed in (6.5) simplify to

$$\kappa_3 = 2(\tau^2 - 3\tau + 3), \quad (7.1)$$

$$\kappa_4 = 2(\tau^3 - 8\tau^2 + 19\tau - 13). \quad (7.2)$$

These follow by repeated application of $t d/dt$ to $\log(e^t - 1)$ before substituting $\mu(\tau) = 2$. In particular, the constraint $\mu = 2$ must *not* be imposed before differentiation.

Define, for $\delta \in \{1/2, 1\}$,

$$J_\delta = \rho - \delta^2 - 2\delta - \frac{1}{12} + 2E(-2\delta), \quad (7.3)$$

where E is given by (6.5) with v, κ_3, κ_4 as above. Numerically,

$$J_1 = 3.3283713212796944340 \dots, \quad J_{1/2} = 6.5286224181582893947 \dots \quad (7.4)$$

Theorem 7.1 (First-corrected tail). *With $T(n)$ defined in (5.7),*

$$D_n = T(n) \left(1 + \frac{J_{\delta_n}}{n} + O(n^{-2}) \right). \quad (7.5)$$

Proof. There are three independent first-order contributions. First, the exact factorial prefactor in (5.6) satisfies

$$\frac{n! g^K \tau^{-n}}{K! \sqrt{2\pi K v}} = T(n) \left(1 - \frac{\delta^2 + 2\delta + 1/12}{n} + O(n^{-2}) \right). \quad (7.6)$$

For example, this follows from the shifted Stirling expansion

$$\log \Gamma(x + \delta + 1) = (x + \delta + \tfrac{1}{2}) \log x - x + \tfrac{1}{2} \log(2\pi) + \frac{\delta(\delta+1)/2 + 1/12}{x} + O(x^{-2}),$$

used with $x = n/2$, together with the expansion of $K^{-1/2}$. Second, the probability in (6.4) contributes

$$1 + \frac{E(-2\delta)}{K} + O(K^{-2}) = 1 + \frac{2E(-2\delta)}{n} + O(n^{-2}).$$

Third, the sum of the omitted terms contributes $1 + \rho/n + O(n^{-2})$ by (5.5). Multiplication gives (7.3) and (7.5). $\square$

This refinement distinguishes two tasks. To approximate $a(n)$ at ordinary relative precision, it is enough to use the Bell expansion. To approximate $B_n - a(n)$, subtracting separately rounded approximations of $a(n)$ and B_n is numerically inappropriate; use the tail formula itself. The latter is an exponentially smaller scale than the error of any fixed algebraic truncation of the Bell expansion.

8 The exact scale of the relative deficit

Combining Theorems 1.2 and 5.1 gives the full leading equivalent

$$\frac{D_n}{B_n} \sim \frac{\rho^{\delta_n} \sqrt{1+r}}{\sqrt{\pi(\tau-1)}} n^{-\delta_n-1/2} \exp\left\{\frac{n}{2} \log n + n \log C - n(r-1+r^{-1}) + 1\right\}. \quad (8.1)$$

The two formulas also yield, if a first correction to this small probability is desired,

$$\frac{D_n}{B_n} = \frac{T(n)}{F(n)} \left(1 + \frac{J_{\delta_n} + P(r)}{n} + O(r^2/n^2)\right). \quad (8.2)$$

In particular,

$$\begin{aligned} \log \frac{D_n}{B_n} &= \frac{n}{2} \log n + n \log C - n(r-1+r^{-1}) \\ &\quad - (\delta_n + \tfrac{1}{2}) \log n + \delta_n \log \rho + \frac{1}{2} \log(1+r) - \frac{1}{2} \log(\pi(\tau-1)) + 1 + O(r/n). \end{aligned} \quad (8.3)$$

On substituting (4.2), this becomes

$$\begin{aligned} \log \frac{D_n}{B_n} &= -\frac{1}{2} n \log n + n \log \log n + (1 + \log C)n \\ &\quad - \frac{n(\log \log n + 1)}{\log n} - \frac{n(\log \log n)^2}{2(\log n)^2} + O\left(\frac{n(\log \log n)^3}{(\log n)^3}\right). \end{aligned} \quad (8.4)$$

Here

$$\log C = 0.06380671444450928117 \dots$$

The parity terms, although essential for a relative equivalent, fit inside the error of the much coarser logarithmic expansion (8.4). In particular,

$$\lim_{n \rightarrow \infty} \frac{\log(1 - a(n)/B_n)}{n \log n} = -\frac{1}{2}. \quad (8.5)$$

The convergence in this last normalized limit is slow, because its next term is $\log \log n / \log n$. For numerical work at moderate n , the Lambert-W expression (8.1) is substantially more useful than retaining only the limiting coefficient $-1/2$.

9 Extension to every fixed positive fraction

The cutoff $n/2$ is not exceptional as far as leading Bell growth is concerned. The following extension also makes the mechanism transparent.

Theorem 9.1 (Fixed-fraction truncation). *Fix $0 < \alpha < 1$ and define*

$$a_\alpha(n) = \sum_{k \leq \lfloor \alpha n \rfloor} S(n, k), \quad K = \lfloor \alpha n \rfloor + 1, \quad \delta = K - \alpha n \in (0, 1].$$

Let $\tau_\alpha > 0$ solve $\tau_\alpha/(1 - e^{-\tau_\alpha}) = 1/\alpha$, and put

$$\begin{aligned} g_\alpha &= e^{\tau_\alpha} - 1, \\ v_\alpha &= \frac{1}{\alpha} \left(1 + \tau_\alpha - \frac{1}{\alpha} \right), \\ C_\alpha &= \frac{e^{\alpha-1}}{\tau_\alpha} \left(\frac{g_\alpha}{\alpha} \right)^\alpha. \end{aligned}$$

Then $a_\alpha(n) \sim B_n$ and

$$B_n - a_\alpha(n) = \frac{(g_\alpha/\alpha)^\delta}{\alpha \sqrt{2\pi v_\alpha}} n^{-\delta-1/2} (C_\alpha n^{1-\alpha})^n (1 + O_\alpha(n^{-1})). \quad (9.1)$$

Moreover,

$$\log \frac{B_n - a_\alpha(n)}{B_n} = -\alpha n \log n + n \log \log n + O_\alpha(n). \quad (9.2)$$

Proof. Use the tilt (6.1) with mean $1/\alpha$. At $k = K$, the offset $n - K/\alpha = -\delta/\alpha$ is bounded. Lemma 6.1 applies uniformly to these offsets. The argument in (6.9) shows that the tail is its first term times $1 + O_\alpha(1/n)$; in fact the next-term contribution is $g_\alpha/(\alpha n) + O_\alpha(n^{-2})$. Stirling's formula gives (9.1). Taking logarithms and using (1.8) for B_n gives (9.2), which proves the asymptotic equivalence. When $\alpha = 1/2$, the constants reduce to those in Theorem 5.1. $\square$

The hypothesis that α is fixed is important. The theorem is not a uniform statement as $\alpha \downarrow 0$. In particular, a cutoff moving near $n/\log n$ lies on the typical block-count scale and requires a different analysis. For irrational fixed α , the fractional offset δ need not converge; retaining it in (9.1) supplies the appropriate uniform prefactor.

10 Numerical checks and reproducibility

10.1 Exact computation and validation

The accompanying Python program computes the Stirling triangle by

$$S(0, 0) = 1, \quad S(n, k) = kS(n-1, k) + S(n-1, k-1), \quad (10.1)$$

with out-of-range values zero. It then sums each row to obtain B_n , the prefix to obtain $a(n)$, and the remaining suffix to obtain D_n . Thus no large-number subtraction is used to form a tiny relative deficit. The recurrence has an immediate combinatorial proof: either the newest element joins one of k blocks or forms a new singleton.

Exact counts were computed through $n = 2001$. All displayed OEIS values for $0 \leq n \leq 25$ agree. The Bell totals were independently checked through $n = 400$ using

$$B_{n+1} = \sum_{j=0}^n \binom{n}{j} B_j.$$

This second recurrence is also listed by NIST [3]. The four cumulant formulas were checked against numerical derivatives of the cumulant generating function at 80 decimal digits. These checks support the algebra and implementation; they are not substitutes for the proofs above. The supplied verification report states exactly which checks were executed. The longer OEIS b-file [2] can be checked separately with the program's optional `-ois-file` argument; it was not used as a full external validation in this run.

The exact algorithm takes $O(N^2)$ arbitrary-precision integer operations and retains one Stirling row at a time, apart from the output and the small validation table. This is an arithmetic-operation count, not a bit-complexity bound: the integers themselves grow with n .

10.2 Leading and first-corrected approximations

Write

$$F_1(n) = F(n)(1 - P(r)/n), \quad T_1(n) = T(n)(1 + J_{\delta_n}/n).$$

Table 1 separates the effect of the cutoff from the error of the Bell approximation. All columns headed by a ratio minus one are *signed* relative errors, computed from exact integer counts.

n	$(B_n - a(n))/B_n$	$F(n)/a(n) - 1$	$F_1(n)/a(n) - 1$
10	2.5439×10^{-1}	3.6567×10^{-1}	3.4195×10^{-1}
20	4.6008×10^{-2}	5.9867×10^{-2}	4.8402×10^{-2}
50	6.5861×10^{-6}	5.5674×10^{-3}	3.8427×10^{-5}
100	7.3158×10^{-16}	3.2252×10^{-3}	8.7765×10^{-6}
200	1.4574×10^{-43}	1.8424×10^{-3}	2.36×10^{-6}
400	1.7691×10^{-116}	1.0391×10^{-3}	6.1994×10^{-7}
1000	3.132×10^{-404}	4.7952×10^{-4}	1.023×10^{-7}
2000	1.4659×10^{-997}	2.6438×10^{-4}	2.5497×10^{-8}

Table 1: The omitted proportion and the relative errors of the Bell approximations. At large n , the error in F or F_1 is overwhelmingly larger than the truncation error.

For instance, at $n = 100$ the omitted proportion is approximately $7.31581910 \times 10^{-16}$, while at $n = 1000$ it is approximately $3.13199596 \times 10^{-404}$. The first-corrected Bell approximation has relative error about 1.0230×10^{-7} at $n = 1000$. Those numbers illustrate the separation of scales: improving the Bell expansion is much more important than accounting for the cutoff when approximating $a(n)$ itself.

n	$n(D_n/S(n, K_n) - 1)$	$T(n)/D_n - 1$	$T_1(n)/D_n - 1$
50	6.8039771	-3.8391×10^{-2}	2.5621×10^{-2}
51	7.4749692	-1.0536×10^{-1}	9.1698×10^{-3}
100	7.3721086	-2.6107×10^{-2}	6.3077×10^{-3}
101	7.7167024	-5.889×10^{-2}	1.9435×10^{-3}
400	7.7380382	-7.8721×10^{-3}	3.8331×10^{-4}
401	7.8241196	-1.5923×10^{-2}	9.8749×10^{-5}
2000	7.8228653	-1.6463×10^{-3}	1.5175×10^{-5}
2001	7.8400188	-3.2484×10^{-3}	3.6707×10^{-6}

Table 2: The endpoint ratio and the two parity-sensitive tail approximations. The second column tends to $\rho = 7.8431072691 \dots$

Table 2 confirms the different parity corrections. These are asymptotic formulas: adding a term need not improve small- n accuracy. The diagnostics file includes small values as well as large ones.

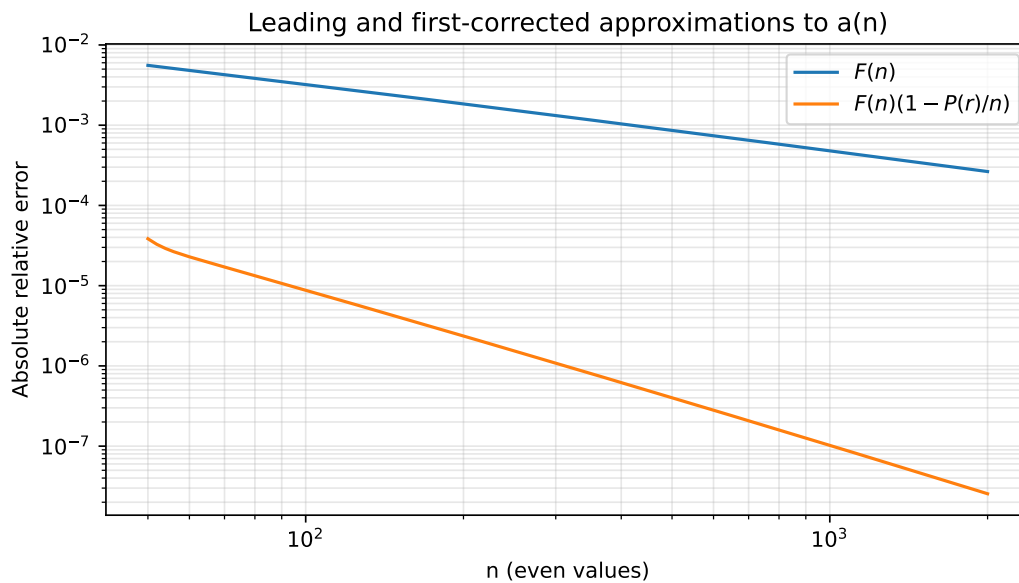


Figure 1: Absolute relative errors against exact $a(n)$ for even $n \geq 50$. The first correction improves the approximation by several orders of magnitude over the plotted range. These are numerical observations; the proved error orders are in Theorem 1.2.

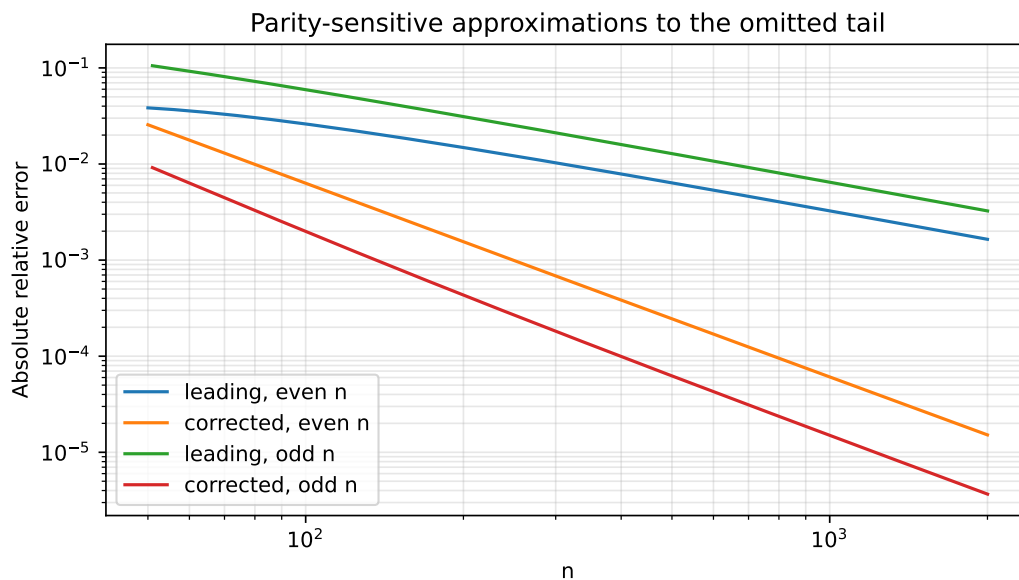


Figure 2: Relative tail errors for the leading and first-corrected formulas, shown separately for even and odd indices. The exact tail is computed as a sum of Stirling numbers, avoiding cancellation.

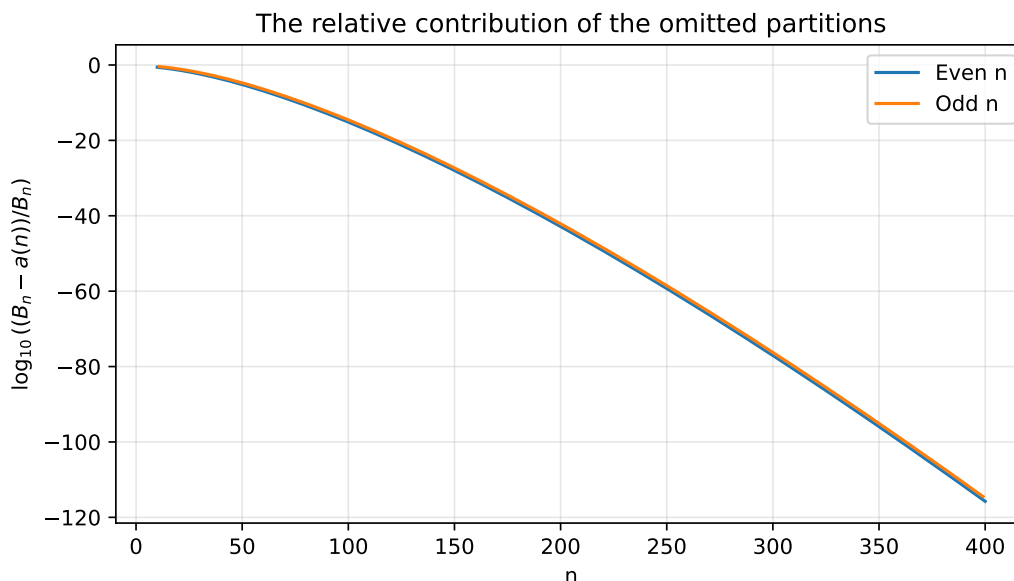


Figure 3: The base-10 logarithm of the omitted proportion. The small parity effect is visible, but does not alter the leading Bell growth.

10.3 Stable evaluation

For a large index, evaluate the logarithm first:

$$\log F(n) = n(r - 1 + r^{-1}) - 1 - \frac{1}{2} \log(1 + r), \quad r = W_0(n). \quad (10.2)$$

Exponentiate only when a numerical value rather than its logarithm is needed. For the deficit, use $\log T(n) - \log F(n)$, or compute D_n/B_n from separate exact integers. Computing $1 - a(n)/B_n$ in fixed precision will eventually return zero even though $D_n > 0$.

The supplied script uses 80-decimal-digit arithmetic for approximations. Its exact integer counts are unaffected by that precision setting. The tail probability can have an exponent below -400 or -900 without requiring 400 or 900 mantissa digits, because it is evaluated directly, not obtained by subtracting a number close to one.

10.4 Archive contents and commands

The archive includes this source and compiled PDF, the Python programs, three figures in PDF and PNG formats, an independently generated exact table for $0 \leq n \leq 1000$, a diagnostic CSV for $2 \leq n \leq 2001$, and a machine-readable verification summary. An OEIS-style formula note is included as a convenient extract; no submission to OEIS has been made.

The mathematical proofs do not depend on external software, a network connection, or any unverified numerical hypothesis. From the archive directory, the calculations and article can be rebuilt with

```
python -m pip install -r requirements.txt
python code/analyze.py --max-n 2001 --plots
python code/make_tables.py
latexmk -pdf -interaction=nonstopmode A277364_asymptotics.tex
```

11 Conclusion

The requested growth rate is fully determined by the classical Bell saddle. The cutoff at $n/2$ is asymptotically far enough to the right that it does not change any fixed algebraic-order relative expansion. The precise leading equivalent is (1.7), and its first correction is (1.6).

The restriction nevertheless has a sharply describable signature on its own scale. The difference $B_n - a(n)$ is controlled by the first omitted Stirling number, with an explicit correction for the rest of the tail. It grows like $(C\sqrt{n})^n$ with a parity-dependent power of n , whereas B_n grows on the much larger $(n/\log n)^n$ scale. Equations (7.5) and (8.2) quantify both effects. These are sequence-specific applications and refinements of classical coefficient asymptotics, not claims of a new general Bell-number theory.

References

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- [3] NIST Digital Library of Mathematical Functions, §26.7: *Set Partitions: Bell Numbers*, particularly (26.7.2), (26.7.5)–(26.7.8). <https://dlmf.nist.gov/26.7>. Accessed 20 September 2026.
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